

Chapter 10: Analysing data

8 lessons · one lesson for each exercise · each lesson is one hour

NSW syllabus outcome

Stage 4 Statistics: MA4-DAT-C-01 classifies and displays data using a variety of graphical representations; MA4-DAT-C-02 analyses simple datasets using measures of centre, range and shape of the data.

CHAPTER

10

10.01 Interpreting graphs

10.02 Misleading graphs

10.03 Dot plots

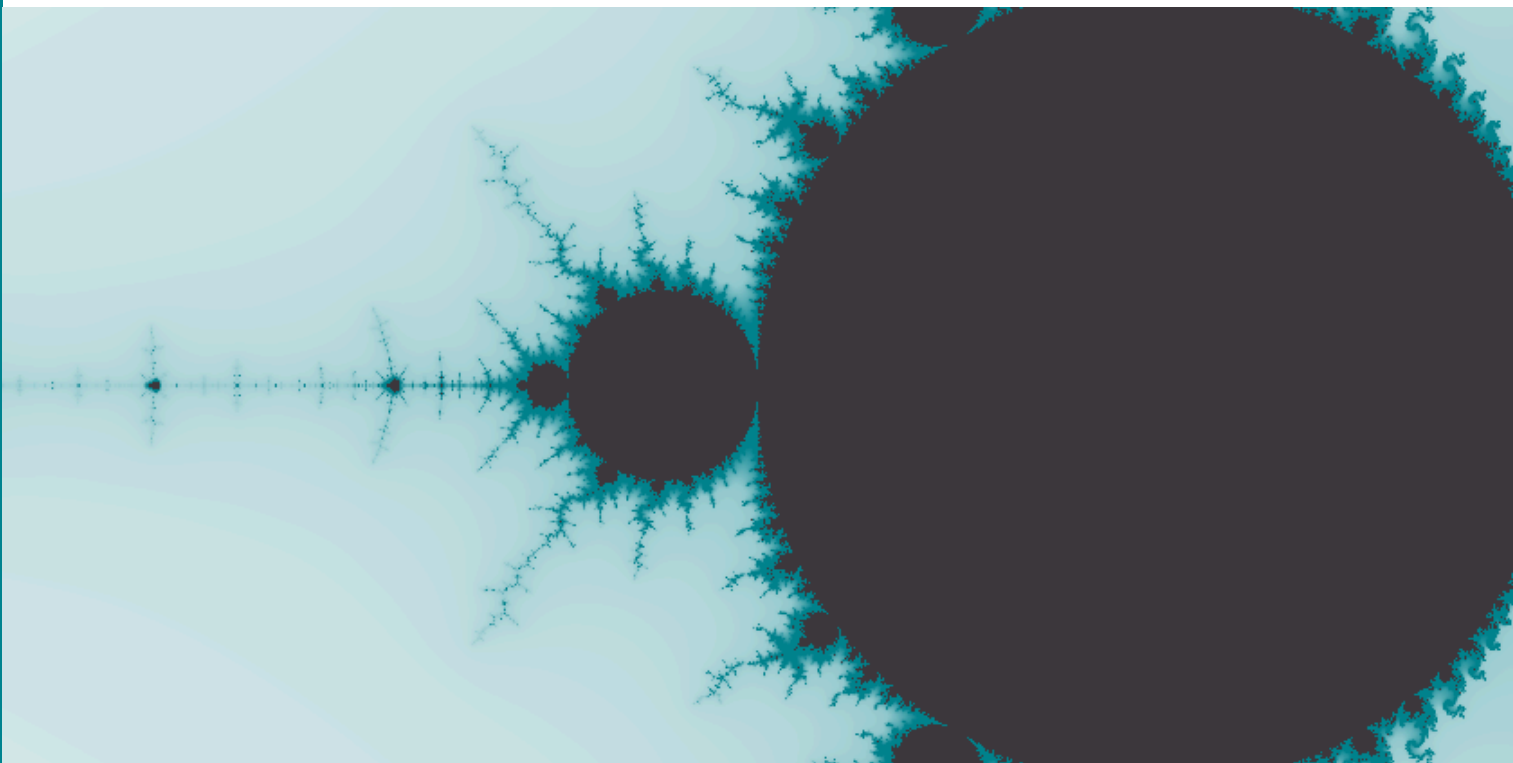
10.04 Stem-and-leaf plots

10.05 Mean and mode

10.06 Median and range

10.07 Analysing dot plots and stem-and-leaf plots

10.08 Comparing data



How to use this booklet

1 Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

2 Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

WE DO **Grey boxes.** Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

YOU DO **Blue boxes.** Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

3 Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

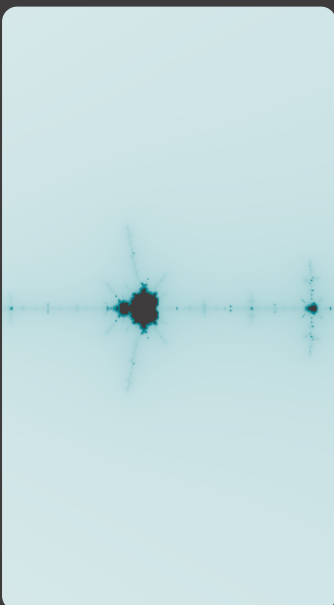
4 Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
10.01	Interpreting graphs	3		☐
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10.04	Stem-and-leaf plots	21		☐
10.05	Mean and mode	27		☐
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In this chapter you will

- interpret picture, column, sector and line graphs
- identify misleading graphs
- read and draw dot plots and stem-and-leaf plots
- find the mean, mode, median and range
- analyse data shown in plots
- compare two sets of data



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
10.01

Interpreting graphs

Date _____

Learning intentions

We are learning to:

- read and interpret picture, sector, column and line graphs

Success criteria

I can:

- ☐ read the title, axes, key and scale first
- ☐ read values from a graph
- ☐ answer questions and explain trends

Key terms Write the definition as your teacher explains it.

Key

Scale

Sector graph

Notes

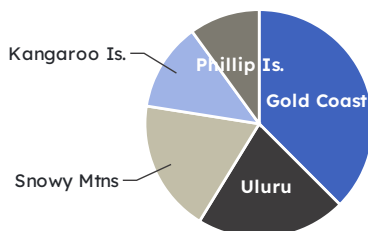
Level 1 · Getting started

Easy · 14 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the sector graph.

Favourite holiday destinations (80 people)



a What is the most popular destination?

Your teacher works this out. Copy every step.

b Which destination did 17 people prefer?

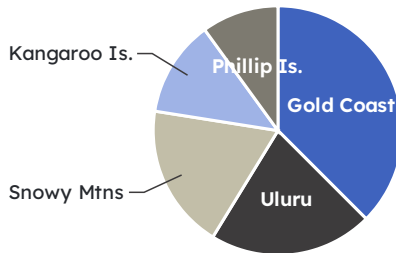
Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Use the sector graph.

4 QUESTIONS

REMEMBER Read the title, labels and key.

Favourite holiday destinations (80 people)



- 1 Most popular destination?
- 2 Which did 17 people prefer?
- 3 Estimate how many chose the Gold Coast.
- 4 True or false: fewer than 10 chose Phillip Island.

YOU DO On your own: **Easy** Set A2. Quick drill: reading scales. A column reaches the line shown. What is its value?

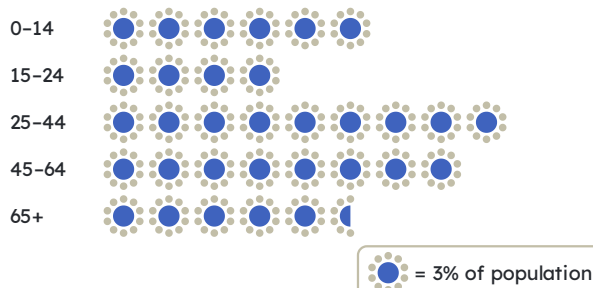
6 QUESTIONS

- 1 halfway between 10 and 20
- 2 2 small steps above 40 (steps of 5)
- 3 just below 30 on a scale of 2s
- 4 3 symbols, key: 1 symbol = 4 people
- 5 $2\frac{1}{2}$ symbols, key: 1 symbol = 10
- 6 a quarter of a circle of 80 people

YOU DO On your own: **Easy** Set B. Use the picture graph.

4 QUESTIONS

Age of Australia's population



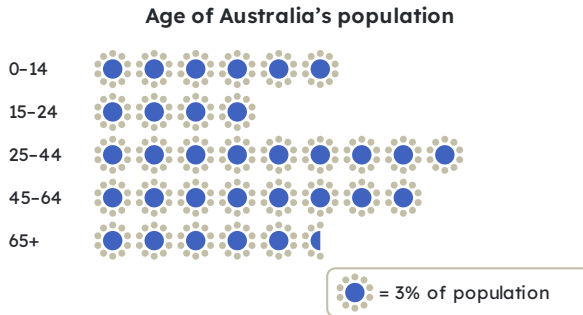
- 1 Smallest age group?
- 2 Percentage aged 15-24?
- 3 Percentage aged 65+?
- 4 Why might 25-44 be the largest group?

Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Use the picture graph.



a Which age group has the smallest percentage?

Your teacher works this out. Copy every step.

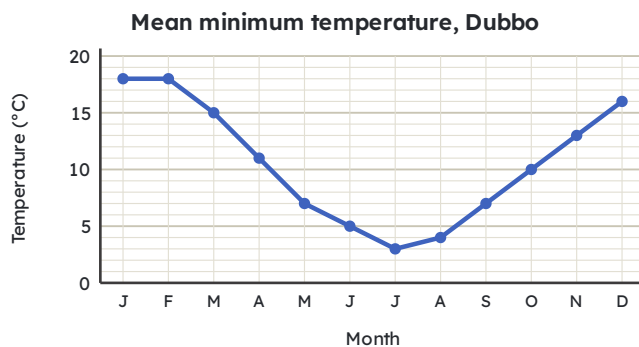
b What percentage are aged 15-24?

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Use the line graph.

6 QUESTIONS

REMEMBER Check the scale on each axis.



1 Highest months and temperature?

2 Two months with the same temperature?

3 Which month has 10°C?

4 Coldest month?

5 Between which 2 months is the smallest drop?

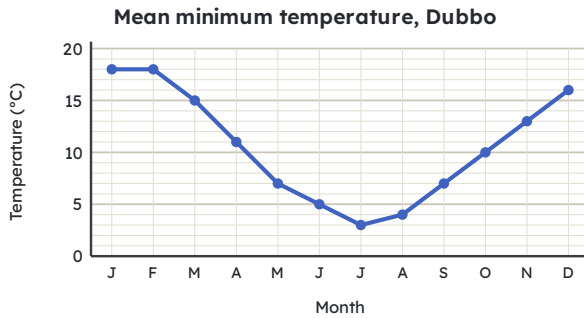
6 Describe the trend from January to July.

Level 3 · Challenge

Challenging · 4 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Use the line graph.



- a** Which months have the highest mean minimum?
What is it?

Your teacher works this out. Copy every step.

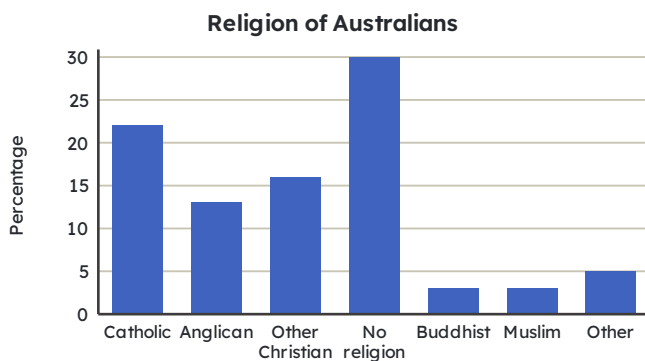
- b** Describe the trend from January to July.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Use the column graph.

4 QUESTIONS

REMEMBER Read values carefully with a ruler.



- 1** What percentage are Catholic?

- 2** Which religion is followed by 13%?

- 3** What is the most common response?

- 4** How might this graph have been different 20 years ago?

Extension · Stretch yourself

Finished Set D? Stretch yourself here.

Worked example

In the sector graph, what angle is the Gold Coast sector?

1. Gold Coast is 30 out of 80 people
2. $\frac{30}{80} \times 360^\circ$

Answer: 135°

YOU DO

On your own: read the worked example, then try these.

- E1** Find the angle of the Uluru sector (1 d.p.).

- E2** Draw a divided bar graph 16 cm long for the holiday data. How long is each part?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

In the sector graph, which destination was least popular?

Level 2

In the line graph, what is the mean minimum in April?

Level 3

In the column graph, how many more % have no religion than are Catholic?

Lesson summary

Missed the lesson? Start here.

Remember

1. Read the title, labels and key.
2. Check the scale on each axis.
3. Read values carefully with a ruler.
4. Describe trends: increasing, decreasing, highest, lowest.

Level 1

Most popular (sector)

1. Largest sector

Answer: Gold Coast

Level 2

15–24 (picture graph)

1. 4 symbols $\times$ 3%

Answer: 12%

Level 3

Highest mean minimum

1. Highest points

Answer: Jan and Feb,
18°C

Extension

Gold Coast angle

1. $30/80 \times 360$

Answer: 135°

My reflection: what did I learn today? Which question was hardest, and why?

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LESSON
10.02

Misleading graphs

Date _____

Learning intentions

We are learning to:

- identify features that make a graph misleading

Success criteria

I can:

- ☐ check whether the vertical axis starts at zero
- ☐ check the scale is even and the graph has labels
- ☐ redraw a misleading graph correctly

Key terms Write the definition as your teacher explains it.

Misleading

Broken axis

Notes

Level 1 · Getting started

Easy · 14 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the line graph.



a What is incorrect about it?

Your teacher works this out. Copy every step.

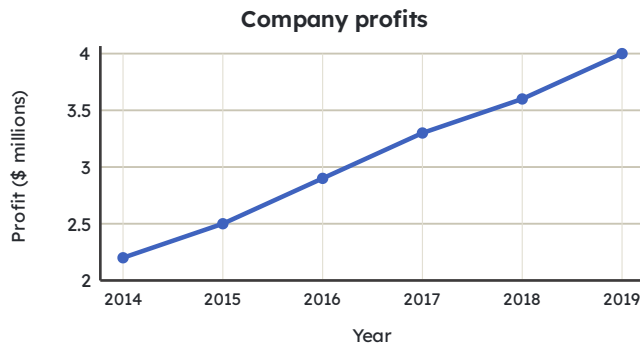
b What misleading impression does it give?

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Use the line graph.

4 QUESTIONS

REMEMBER Check the vertical axis starts at 0 (or shows a break).



- 1 What is incorrect about the graph?
- 2 What impression does it give?
- 3 What was the profit in 2014?
- 4 By how much did profit grow from 2014 to 2019?

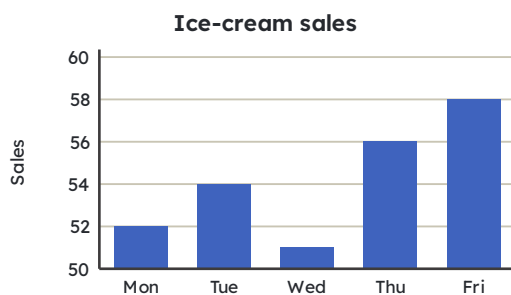
YOU DO On your own: **Easy** Set A2. Quick drill: misleading or fair? Write M or F.

6 QUESTIONS

- 1 The vertical axis starts at 50
- 2 The scale goes 0, 10, 20, 30
- 3 The scale goes 0, 10, 20, 50, 100
- 4 There is no title or labels
- 5 A picture is doubled in height and width
- 6 The axis starts at 0 with a clear key

YOU DO On your own: **Easy** Set B. Use the column graph.

4 QUESTIONS



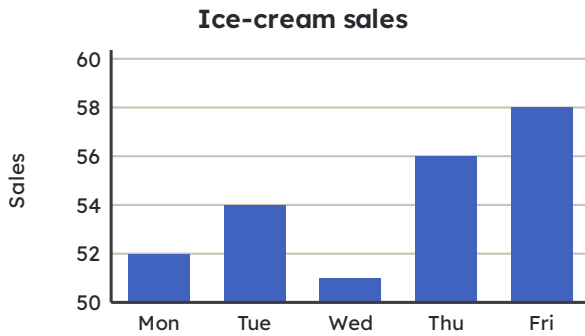
- 1 Where does the vertical axis start?
- 2 Wednesday's sales?
- 3 Friday's sales?
- 4 Does Friday have double Wednesday's sales?

Level 2 · Building up

Medium · 1 question

WE DO With your teacher: copy what they write on the board.

Example 2 Use the column graph.



a Thursday sales look double Wednesday's. Is that true?

Your teacher works this out. Copy every step.

b What are the actual sales on Wednesday and Thursday?

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Redraw the company profits graph correctly, starting the axis at 0.

1 QUESTION

REMEMBER Check the scale goes up in equal steps.

1 Profits 2014–2019: 2.2, 2.5, 2.9, 3.3, 3.6, 4.0 (\$ millions)

Redraw here

Level 3 · Challenge

Challenging · 4 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Think about misleading graphs.

a Name 3 other ways a graph can mislead.

Your teacher works this out. Copy every step.

b Why would a company want a misleading graph?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Think it through.

4 QUESTIONS

REMEMBER Check there is a title and labelled axes.

1 A graph uses pictures of bags of money; the 2019 bag is twice as tall and twice as wide as the 2014 bag. Why is this misleading?

3 A column graph has a scale 0, 10, 20, 50, 100. What is wrong?

2 A graph has no title and no labels on the axes. Why is this a problem?

4 Find a graph in a newspaper or online. Is it misleading? How?

Extension · Stretch yourself

Finished Set D? Stretch yourself here.

Worked example

Profit grew from \$2.2 million to \$4.0 million. By what percentage did it grow?

1. Increase = 1.8 million
2. $\frac{1.8}{2.2} \times 100\%$

Answer: $\approx 81.8\%$

YOU DO

On your own: read the worked example, then try these.

- E1 On the misleading graph, the 2019 point looks about 8 times higher than 2014. Explain why.

- E2 Draw a graph of the same data that makes growth look very small.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

Give one reason a graph might be misleading.

Level 2

A column graph's axis starts at 90. Why is that a problem?

Level 3

How should the company profits graph be fixed?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Check the vertical axis starts at 0 (or shows a break).
 2. Check the scale goes up in equal steps.
 3. Check there is a title and labelled axes.
 4. Watch for pictures that change in 2 directions.

Level 1

Profit graph

1. Axis starts at 2

Answer: growth looks bigger

Level 2

Sales Wed vs Thu

1. 51 vs 56

Answer: not double

Level 3

Uneven scale 0, 10, 20, 50

1. Steps not equal

Answer: misleading

Extension

Growth %

1. $1.8 \div 2.2$

Answer: $\approx 81.8\%$

My reflection: what did I learn today? Which question was hardest, and why?

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LESSON
10.03

Dot plots

Date _____

Learning intentions

We are learning to:

- read and draw dot plots

Success criteria

I can:

- ☐ read the frequency of each value
- ☐ find the mode and any outlier
- ☐ draw a dot plot from data

Key terms Write the definition as your teacher explains it.

Dot plot

Outlier

Mode

Notes

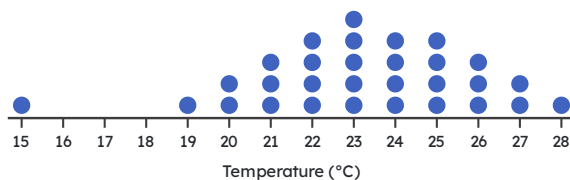
Level 1 · Getting started

Easy · 14 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the dot plot.

Maximum temperatures at Bega in April



a What is the mode?

Your teacher works this out. Copy every step.

b What is the outlier?

Your teacher works this out. Copy every step.

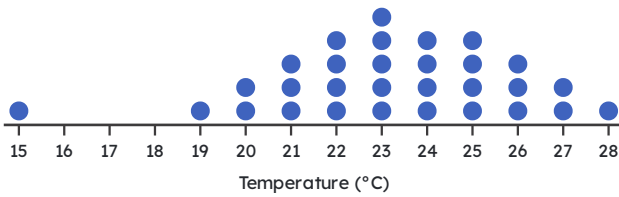
YOU DO

On your own: **Easy** Set A. Use the dot plot.

4 QUESTIONS

REMEMBER Each dot is one piece of data.

Maximum temperatures at Bega in April



1 Mode?

2 Outlier?

3 Days at 25°C?

4 Fraction of days below 20°C?

YOU DO

On your own: **Easy** Set A2. Quick drill: find the mode.

6 QUESTIONS

1 2, 3, 3, 4, 5

2 7, 8, 8, 8, 9, 9

3 1, 1, 2, 2, 2, 3

4 10, 12, 12, 15

5 5, 6, 6, 7, 7, 7

6 0, 0, 1, 3, 3, 3, 4

YOU DO

On your own: **Easy** Set B. Use the dot plot.

4 QUESTIONS

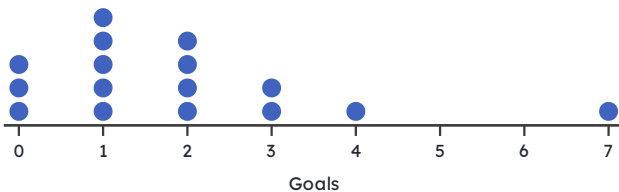
1 How many games?

2 Mode?

3 Outlier?

4 How many games had 2 or more goals?

Goals scored per game



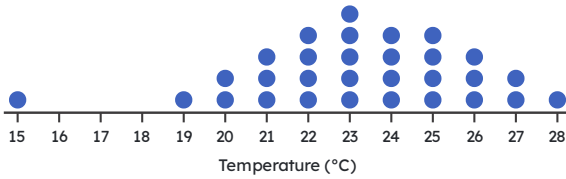
Level 2 · Building up

Medium · 2 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Use the dot plot.

Maximum temperatures at Bega in April



a On how many days was it 25°C?

Your teacher works this out. Copy every step.

b On what fraction of days was it below 20°C?

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Draw a dot plot for each set of data.

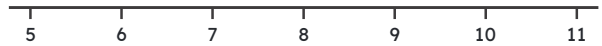
2 QUESTIONS

REMEMBER Mode: the tallest column.

1 Number of pets: 0, 1, 1, 2, 0, 3, 1, 2, 1, 0, 4, 1



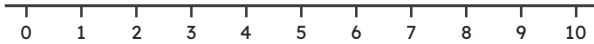
2 Shoe sizes: 7, 8, 8, 9, 7, 10, 8, 9, 8, 6, 8, 9





Level 3 · Challenge

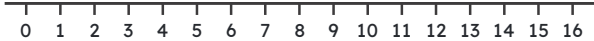
Challenging · 2 questions

WE DO With your teacher: copy what they write on the board.**Example 3** Draw a dot plot, then describe.**a** Draw a dot plot: 2, 3, 3, 4, 4, 4, 5, 6, 6, 9**b** Describe the shape of the Set A plot.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Draw and analyse.

2 QUESTIONS

REMEMBER Outlier: far away from the rest.**1** Minutes late: 0, 2, 1, 0, 3, 0, 15, 1, 2, 0, 1, 0. Draw a dot plot.**2** For Question 1, find the mode and the outlier. How does the outlier affect the data?



Extension · Stretch yourself

Finished Set D? Stretch yourself here.



Worked example

What percentage of April days at Bega were 25°C or more?

- Count: $25 (4) + 26 (3) + 27 (2) + 28 (1) = 10$
- $\frac{10}{30} \times 100\%$

Answer: 33.3%

YOU DO

On your own: read the worked example, then try these.

- E1 What percentage of games had no goals? (Set B)

- E2 Add a game with 3 goals to Set B. What is the new mode?

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

What is a dot plot?

Level 2

In Set B, what is the mode?

Level 3

Draw a dot plot for 1, 2, 2, 3, 3, 3, 8. Which value is the outlier?

Lesson summary

Missed the lesson? Start here.

Remember

- Each dot is one piece of data.
- Mode: the tallest column.
- Outlier: far away from the rest.
- Draw: a number line with an even scale, one dot per value.

Level 1

Mode of Bega plot

- Tallest column

Answer: 23°C

Level 2

Outlier

- Far from the rest

Answer: 15°C

Level 3

Below 20°C

- 2 days out of 30

Answer: $\frac{1}{15}$

Extension

25°C or more

- 10 of 30 days

Answer: 33.3%

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
10.04

Stem-and-leaf plots

Date _____

Learning intentions

We are learning to:

- read and draw ordered stem-and-leaf plots

Success criteria

I can:

- ☐ read values from a stem-and-leaf plot using the key
- ☐ order the leaves in each row
- ☐ draw a stem-and-leaf plot from data

Key terms Write the definition as your teacher explains it.

Stem

Leaf

Key

Notes

Level 1 · Getting started

Easy · 14 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the stem-and-leaf plot.

Masses of football players (kg), unordered

Stem	Leaf
7	9 5 8 1 6 9 4
8	2 5 0 0 8 0 3 2 4
9	0 8 4 2 2 0
10	5 6 4

Key: 7 | 9 = 79 kg

a How many players are there?

Your teacher works this out. Copy every step.

b What is the lowest mass?

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Use the stem-and-leaf plot.

4 QUESTIONS

REMEMBER The stem is the tens digit; the leaf is the ones digit.

Masses of football players (kg), unordered

Stem	Leaf
7	9 5 8 1 6 9 4
8	2 5 0 0 8 0 3 2 4
9	0 8 4 2 2 0
10	5 6 4

Key: 7 | 9 = 79 kg

1 How many players?

2 Lowest mass?

3 Most common mass?

4 Highest mass?

YOU DO On your own: **Easy** Set A2. Quick drill: write the value (key 3 | 4 = 34).

6 QUESTIONS

1 2 | 7

2 5 | 0

3 1 | 9

4 8 | 3

5 10 | 4

6 6 | 6

YOU DO On your own: **Easy** Set B. Use the stem-and-leaf plot.

4 QUESTIONS

1 How many scores?

2 Lowest score?

3 Highest score?

4 Mode?

Test scores

Stem	Leaf
4	5
5	0 0 1 2
6	1 3 4 6 6 6 7 8
7	4 5

Key: 5 | 2 = 52

Level 2 · Building up

Medium · 2 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Use the stem-and-leaf plot.

Masses of football players (kg), unordered

Stem	Leaf
7	9 5 8 1 6 9 4
8	2 5 0 0 8 0 3 2 4
9	0 8 4 2 2 0
10	5 6 4

Key: 7 | 9 = 79 kg

a Which mass occurs most often?

Your teacher works this out. Copy every step.

b What is the highest mass?

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Draw an ordered stem-and-leaf plot.

2 QUESTIONS

REMEMBER Always include a key, e.g. 7 | 9 = 79.

1 23, 35, 41, 28, 32, 45, 39, 27, 30, 44, 36, 21

Stem	Leaf
2	
3	
4	

Key: ____ | ____ = _____

2 67, 82, 75, 91, 78, 64, 88, 70, 73, 85, 79, 68

Stem	Leaf
6	
7	
8	
9	

Key: ____ | ____ = _____



Level 3 · Challenge

Challenging · 2 questions

WE DO

With your teacher: copy what they write on the board.

Example 3

Draw an ordered plot, then explain.

a Draw an ordered stem-and-leaf plot for Set A.

Stem	Leaf
7	
8	
9	
10	

Key: ____ | ____ = _____

b Why do we order the leaves?

Your teacher works this out. Copy every step.

YOU DO

On your own: **Challenging** Set D. Draw and analyse.

2 QUESTIONS

REMEMBER Order the leaves from smallest to largest.

1 Order the Set A plot (masses of players).

Stem	Leaf
7	
8	
9	
10	

Key: ____ | ____ = _____

2 For the ordered plot, how many players weigh 90 kg or more? What fraction is that?

Extension · Stretch yourself

Finished Set D? Stretch yourself here.

Worked example

Write the data 3.4, 3.7, 4.1, 4.5, 4.5, 5.0, 5.2 as a stem-and-leaf plot.

1. Stems are the whole numbers, leaves the tenths
2. Key: $3 \mid 4 = 3.4$

Answer: $3 \mid 4 \ 7; 4 \mid 1 \ 5 \ 5; 5 \mid 0 \ 2$

YOU DO

On your own: read the worked example, then try these.

- E1** Draw a stem-and-leaf plot for 105, 112, 118, 121, 125, 130 (stems 10, 11, 12, 13).

- E2** What would a key of $2 \mid 5 = 0.25$ mean?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

In the key $6 \mid 3 = 63$, what is the stem?

Level 2

Write the values in the row $4 \mid 0 \ 2 \ 7$.

Level 3

Draw an ordered stem-and-leaf plot for 12, 25, 17, 21, 14, 29.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. The stem is the tens digit; the leaf is the ones digit.
 2. Always include a key, e.g. $7 | 9 = 79$.
 3. Order the leaves from smallest to largest.
 4. Line the leaves up in columns so the plot shows the shape.

Level 1

Players

1. Count the leaves

Answer: 25

Level 2

Lowest mass

1. First leaf of first stem

Answer: 71 kg

Level 3

Ordered plot

1. Sort each row

Answer: $7 | 1\ 4\ 5\ 6\ 8\ 9\ 9$;
 $8 | 0\ 0\ 0\ 2\ 2\ 3\ 4\ 5\ 8$; $9 | 0$
 $0\ 2\ 2\ 4\ 8$; $10 | 4\ 5\ 6$

Extension

Decimals

1. Stem = whole number

Answer: $3 | 4\ 7$; $4 | 1\ 5\ 5$;
 $5 | 0\ 2$

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
10.05

Mean and mode

Date _____

Learning intentions

We are learning to:

- find the mean and mode of a set of data

Success criteria

I can:

- ☐ find the mean: add the values, divide by how many
- ☐ find the mode: the most common value
- ☐ round the mean to 2 decimal places

Key terms Write the definition as your teacher explains it.

Mean

Mode

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Find the mean and the mode.

a Find the mean of 4, 3, 2, 5, 6, 4, 4.

b Find the mode of 4, 3, 2, 5, 6, 4, 4.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Find the mode.

12 QUESTIONS

REMEMBER Mean = sum of values ÷ number of values.

1 4, 3, 2, 5, 6, 4, 4

2 6, 12, 11, 12, 10, 6, 6, 10, 6

3 8, 4, 1, 1, 4, 1, 3, 6

4 3, 7, 7, 2, 9, 3, 7

5 10, 12, 15, 12, 10, 12

6 red, blue, red, green, blue, red

Questions 7–12: find the mean.

7 2, 4, 6

8 1, 2, 3, 6

9 5, 5, 5, 5

10 10, 20, 30

11 3, 4, 8

12 0, 6, 6, 8

YOU DO On your own: **Easy** Set B. Find the mean.

6 QUESTIONS

1 2, 4, 6, 8

4 5, 5, 5, 5

2 4, 3, 2, 5, 6, 4, 4

5 1, 2, 3, 4, 5, 6

3 10, 20, 30, 40, 50

6 7, 9, 11

Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Find the mean and the mode.

a Find the mean of 6, 12, 11, 12, 10, 6, 6, 10, 6.

Your teacher works this out. Copy every step.

b Find the mode of 6, 12, 11, 12, 10, 6, 6, 10, 6.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Find the mean (2 d.p.) and mode.

6 QUESTIONS

REMEMBER Mode = the value that occurs most often.

1 6, 12, 11, 12, 10, 6, 6, 10, 6

3 12, 15, 11, 15, 17, 14

5 45, 52, 38, 52, 61, 49, 52

2 8, 4, 1, 1, 4, 1, 3, 6

4 3.5, 4.2, 3.5, 5.1

6 0, 0, 1, 2, 2, 2, 3, 5

Level 3 · Challenge

Challenging · 4 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Find the mean and mode, then solve.

a Find the mean and mode of 8, 4, 1, 1, 4, 1, 3, 6.

Your teacher works this out. Copy every step.

b The mean of 4 numbers is 7. Three are 5, 8 and 9. Find the fourth.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Problems.

4 QUESTIONS

REMEMBER There can be more than one mode, or none.

1 The mean of 4 numbers is 7. Three are 5, 8 and 9. Find the fourth.

3 Write 5 numbers with a mean of 10 and a mode of 8.

2 Sam's test marks are 72, 85, 68 and 91. What must he score on the next test for a mean of 80?

4 Can a data set have more than one mode? Give an example.

Extension · Stretch yourself

Finished Set D? Stretch yourself here.

Worked example

A class of 25 has a mean mark of 64. A new student scores 90. Find the new mean.

1. Total = $25 \times 64 = 1600$
2. New total 1690, 26 students
3. $1690 \div 26$

Answer: 65

YOU DO

On your own: read the worked example, then try these.

- E1 The mean of 6 numbers is 12. One number, 20, is removed. Find the new mean.

- E2 Explain why the mean does not have to be one of the data values.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

Find the mode: 5, 8, 5, 3, 5, 8.

Level 2

Find the mean: 4, 6, 8, 10, 12.

Level 3

Find the mean (2 d.p.): 3, 7, 7, 9.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Mean = sum of values $\div$ number of values.
 2. Mode = the value that occurs most often.
 3. There can be more than one mode, or none.
 4. Round the mean to 2 decimal places.

Level 1

4, 3, 2, 5, 6, 4, 4

1. sum 28, 7 values

Answer: mean 4

Level 2

4, 3, 2, 5, 6, 4, 4

1. 4 occurs 3 times

Answer: mode 4

Level 3

8, 4, 1, 1, 4, 1, 3, 6

1. sum 28 $\div$ 8

Answer: mean 3.5,
mode 1

Extension

New student

1. $1690 \div 26$

Answer: 65

My reflection: what did I learn today? Which question was hardest, and why?

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LESSON
10.06

Median and range

Date _____

Learning intentions

We are learning to:

- find the median and range of a set of data

Success criteria

I can:

- ☐ order the data before finding the median
- ☐ find the middle value, or the mean of the two middle values
- ☐ find the range: highest – lowest

Key terms Write the definition as your teacher explains it.

Median

Range

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Find the median and the range.

a Find the median of 4, 3, 2, 5, 6, 4, 4.

b Find the range of 4, 3, 2, 5, 6, 4, 4.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Find the range.

12 QUESTIONS

REMEMBER Order the data first.

1 4, 3, 2, 5, 6, 4, 4

2 6, 12, 11, 12, 10, 6, 6, 10, 6

3 8, 4, 1, 1, 4, 1, 3, 6

4 15, 3, 22, 9

5 104, 97, 110, 99

6 -3, 5, 2, -1

Questions 7–12: find the median.

7 1, 3, 5

8 2, 4, 6, 8

9 9, 3, 7

10 5, 1, 4, 2

11 10, 20, 30, 40, 50

12 6, 6, 7, 9

YOU DO On your own: **Easy** Set B. Find the median.

6 QUESTIONS

1 3, 5, 8, 9, 12

4 10, 7, 3, 9, 1

2 4, 3, 2, 5, 6, 4, 4

5 4, 4, 5, 7, 9, 10

3 2, 4, 6, 8

6 20, 15, 30, 25

Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Find the median and the range.

a Find the median of 6, 12, 11, 12, 10, 6, 6, 10, 6.

Your teacher works this out. Copy every step.

b Find the range of 6, 12, 11, 12, 10, 6, 6, 10, 6.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Find the median and range.

6 QUESTIONS

REMEMBER Median = the middle value.

1 6, 12, 11, 12, 10, 6, 6, 10, 6

3 12, 15, 11, 15, 17, 14

5 45, 52, 38, 52, 61, 49, 52

2 8, 4, 1, 1, 4, 1, 3, 6

4 3.5, 4.2, 3.5, 5.1

6 100, 2, 3, 4, 5



Level 3 · Challenge

Challenging · 4 questions

WE DO

With your teacher: copy what they write on the board.

Example 3

Find the median and range, then compare.

- a** Find the median and range of 8, 4, 1, 1, 4, 1, 3, 6.

Your teacher works this out. Copy every step.

- b** Which is affected more by an outlier: the median or the mean?

Your teacher works this out. Copy every step.

YOU DO

On your own: **Challenging** Set D. Problems.

4 QUESTIONS

REMEMBER Even number of values: median = mean of the middle two.

- 1** Find the mean, median, mode and range of 2, 3, 3, 4, 5, 25.

- 3** Write 5 numbers with median 6 and range 10.

- 2** Which measure (mean or median) best describes the data in Question 1? Why?

- 4** The range of a set is 12 and the lowest value is 45. What is the highest?

Extension · Stretch yourself

Finished Set D? Stretch yourself here.

Worked example

Five numbers have mean 6, median 5, mode 3 and range 8. Find them.

1. Mode 3 and median 5: 3, 3, 5, ?, ?
2. Sum = 30, so the last two add to 19
3. Range 8: largest = $3 + 8 = 11$, so the other is 8

Answer: 3, 3, 5, 8, 11

YOU DO

On your own: read the worked example, then try these.

- E1** Four numbers have median 7, mode 9 and range 6. Find them (hint: 9 appears twice).

- E2** Why might a company report the median salary instead of the mean?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the median: 6, 2, 9, 4, 7.

Level 2

Find the range: 18, 5, 12, 30.

Level 3

Find the median: 3, 8, 10, 11.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Order the data first.
 2. Median = the middle value.
 3. Even number of values: median = mean of the middle two.
 4. Range = highest – lowest.

Level 1

4, 3, 2, 5, 6, 4, 4

1. Ordered: 2, 3, 4, 4, 4, 5, 6

Answer: median 4

Level 2

6, 12, 11, 12, 10, 6, 6, 10, 6

1. Ordered: 6, 6, 6, 6, 10, 10, 11, 12, 12

Answer: median 10

Level 3

8, 4, 1, 1, 4, 1, 3, 6

1. $8 - 1$

Answer: range 7

Extension

mean 6, median 5,
mode 3, range 8

1. Build the list

Answer: 3, 3, 5, 8, 11

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
10.07

Analysing dot plots and stem-and-leaf plots

Date _____

Learning intentions

We are learning to:

- find the range, median, mode and mean from a dot plot or stem-and-leaf plot

Success criteria

I can:

- ☐ list or count the data values from the plot
- ☐ use the order of the plot to find the median
- ☐ calculate the mean and range

Key terms Write the definition as your teacher explains it.

Frequency

Cluster

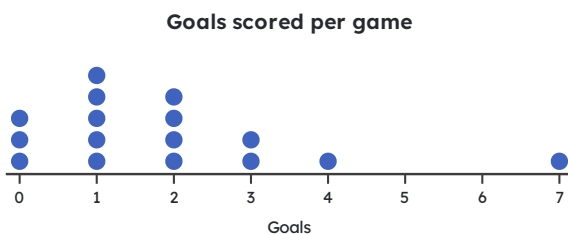
Notes

Level 1 · Getting started

Easy · 14 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the dot plot.



a Find the range.

Your teacher works this out. Copy every step.

b Find the median.

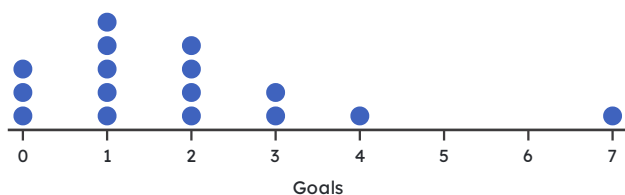
Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Use the dot plot.

4 QUESTIONS

REMEMBER Count the values (n).

Goals scored per game



1 Range

2 Median

3 Mode

4 Mean (2 d.p.)

YOU DO On your own: **Easy** Set A2. Quick drill: find the range.

6 QUESTIONS

1 3, 9, 5

2 12, 4, 20

3 7, 7, 7

4 15, 30, 25, 10

5 0, 8, 3

6 41, 56, 38

YOU DO On your own: **Easy** Set B. Use the stem-and-leaf plot.

4 QUESTIONS

Test scores

Stem	Leaf
4	5
5	0 0 1 2
6	1 3 4 6 6 6 7 8
7	4 5

Key: 5 | 2 = 52

1 Range

2 Median

3 Mode

4 Mean (2 d.p.)

Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Use the stem-and-leaf plot.

Test scores	
Stem	Leaf
4	5
5	0 0 1 2
6	1 3 4 6 6 6 7 8
7	4 5

Key: 5 | 2 = 52

a Find the mode.

Your teacher works this out. Copy every step.

b Find the median.

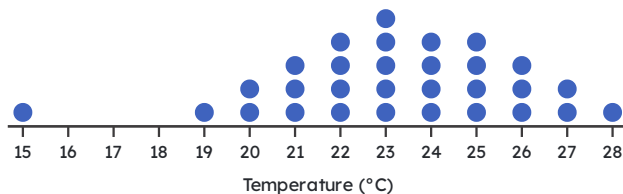
Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Use the dot plot of Bega temperatures.

6 QUESTIONS

REMEMBER Range: highest – lowest.

Maximum temperatures at Bega in April



1 Range

2 Median

3 Mode

4 Mean (2 d.p.)

5 Mean without the outlier (2 d.p.)

6 Which measure changed most when the outlier was removed?

Level 3 · Challenge

Challenging · 4 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Use the stem-and-leaf plot.

Test scores	
Stem	Leaf
4	5
5	0 0 1 2
6	1 3 4 6 6 6 7 8
7	4 5

Key: 5 | 2 = 52

a Set B: find the mean (2 d.p.).

Your teacher works this out. Copy every step.

b Describe where the data in Set B clusters.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Use the stem-and-leaf plot of player masses (ordered).

4 QUESTIONS

REMEMBER Median: the value in position $(n + 1) \div 2$.

Masses of players (kg)	
Stem	Leaf
7	1 4 5 6 8 9 9
8	0 0 0 2 2 3 4 5 8
9	0 0 2 2 4 8
10	4 5 6

Key: 7 | 9 = 79 kg

1 Range

2 Median

3 Mode

4 Mean (2 d.p.)



Extension · Stretch yourself

Finished Set D? Stretch yourself here.



Worked example

One more player of mass 120 kg joins the team.
How do the mean and median change?

1. Old mean 85.88, median 83
2. New mean 87.19, median 83.5

Answer: the mean rises more than the median

YOU DO

On your own: read the worked example, then try these.

- E1** Which measure would you use to describe the typical player's mass? Why?

- E2** Remove the 7-goal game from the dot plot. Find the new mean.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

A dot plot has dots at 1, 2, 2, 3, 3, 3.
Find the mode.

Level 2

Find the median of 5 | 1 3 4 7 (key 5 | 1 = 51).

Level 3

Find the range of 2 | 3 8, 3 | 0 5, 4 | 1
(key 2 | 3 = 23).

Lesson summary

Missed the lesson? Start here.

Remember

1. Count the values (n).
2. Range: highest – lowest.
3. Median: the value in position $(n + 1) \div 2$.
4. Mean: add all values (count dots $\times$ values) $\div n$.

Level 1

Goals: range

1. $7 - 0$

Answer: 7

Level 2

Goals: median

1. $n = 16$

Answer: 1.5

Level 3

Scores: mean

1. sum $918 \div 15$

Answer: 61.2

Extension

Add a 120 kg player

1. Mean moves more

Answer: mean up more

My reflection: what did I learn today? Which question was hardest, and why?

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LESSON
10.08

Comparing data

Date _____

Learning intentions

We are learning to:

- compare two sets of data using plots and statistics

Success criteria

I can:

- ☐ draw a back-to-back stem-and-leaf plot
- ☐ find the mean, median, mode and range for each group
- ☐ compare the groups and decide which is more consistent

Key terms Write the definition as your teacher explains it.

Back-to-back

Consistent

Notes

Level 1 · Getting started

Easy · 14 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the back-to-back plot.

Girls	Stem	Boys
8	1	0 7
9 8	2	8 9
9 7 5	3	0 8
8	4	0
9 5 3 2 1	5	0 5 8
9 6 0	6	2 6
6 5 3	7	2 3 3 4 5
3	8	5 8
4	9	3

Key: 3 | 5 = 35

a What does the plot show?

Your teacher works this out. Copy every step.

b What was the highest mark? Who scored it?

Your teacher works this out. Copy every step.

YOU DO

On your own: **Easy** Set A. Use the back-to-back plot of assignment marks.

4 QUESTIONS

REMEMBER Put the two groups on the two sides of the stems.

Girls	Stem	Boys
8	1	0 7
9 8	2	8 9
9 7 5	3	0 8
8	4	0
9 5 3 2 1	5	0 5 8
9 6 0	6	2 6
6 5 3	7	2 3 3 4 5
3	8	5 8
4	9	3

Key: 3 | 5 = 35

- Highest mark? Boy or girl?
- Lowest mark? Boy or girl?
- How many students scored in the 70s?
- How many girls scored 50 or more?

YOU DO

On your own: **Easy** Set A2. Quick drill: which group is more consistent (smaller range)?

4 QUESTIONS

- A: range 12, B: range 30

- A: range 25, B: range 9

- A: 40 to 60, B: 20 to 90

- A: 5 to 50, B: 30 to 45

YOU DO

On your own: **Easy** Set B. Find each statistic.

6 QUESTIONS

- Girls: mean

- Boys: mean

- Girls: median

- Boys: median

- Girls: range

- Boys: range

Level 2 · Building up

Medium · 2 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Use the back-to-back plot.

Girls	Stem	Boys
8	1	0 7
9 8	2	8 9
9 7 5	3	0 8
8	4	0
9 5 3 2 1	5	0 5 8
9 6 0	6	2 6
6 5 3	7	2 3 3 4 5
3	8	5 8
4	9	3

Key: 3 | 5 = 35

a Find the median mark for the girls.

Your teacher works this out. Copy every step.

b Find the range for the boys.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Draw a back-to-back stem-and-leaf plot.

2 QUESTIONS

REMEMBER Leaves get larger moving away from the stem.

1 Class A: 45, 52, 58, 61, 63, 67, 70, 74. Class B: 38, 49, 55, 56, 60, 72, 81, 88.

Class A	Stem	Class B
	3	
	4	
	5	
	6	
	7	
	8	

Key: ____ | ____ = _____

2 For Question 1, which class did better? Use the median and range.

Level 3 · Challenge

Challenging · 4 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Compare the girls and the boys.

Girls	Stem	Boys
8	1	0 7
9 8	2	8 9
9 7 5	3	0 8
8	4	0
9 5 3 2 1	5	0 5 8
9 6 0	6	2 6
6 5 3	7	2 3 3 4 5
3	8	5 8
4	9	3

Key: 3 | 5 = 35

a Compare the girls and the boys.

Your teacher works this out. Copy every step.

b Which group is more consistent? Why?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Compare and explain.

4 QUESTIONS

REMEMBER Compare centre: mean or median.

1 Comment on the differences between the girls and the boys on the assignment.

3 Which measure (mean or median) would you use to compare these groups? Why?

2 Which group was more consistent? Give reasons.

4 Write 2 questions you could investigate by comparing two groups at school.



Extension · Stretch yourself

Finished Set D? Stretch yourself here.



Worked example

Two basketballers' points in 5 games: Ana 12, 15, 14, 13, 16; Bo 5, 25, 8, 22, 10. Who is more consistent?

1. Both have mean 14
2. Ana range 4; Bo range 20

Answer: Ana (smaller range)

YOU DO

On your own: read the worked example, then try these.

- E1 Who would you pick for a game where you need at least 12 points? Why?

- E2 Make up two data sets with the same mean but very different ranges.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

What is a back-to-back stem-and-leaf plot used for?

Level 2

Group A range 12, group B range 30. Which is more consistent?

Level 3

Group A median 65, group B median 58. Which did better?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Put the two groups on the two sides of the stems.
 2. Leaves get larger moving away from the stem.
 3. Compare centre: mean or median.
 4. Compare spread: a smaller range means more consistent.

Level 1

Highest mark

1. Largest leaf

Answer: 94, a girl

Level 2

Girls median

1. Mean of the 10th and 11th

Answer: 54

Level 3

Boys range

1. $93 - 10$

Answer: 83

Extension

Ana vs Bo

1. Same mean, compare range

Answer: Ana

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.